

HydroSense Kenya

Scientific Computing for Intelligent Irrigation Management

Kenneth Mweu | SCT 211-0383/2024 · Lecturer: Dr. Nderu



Problem Statement

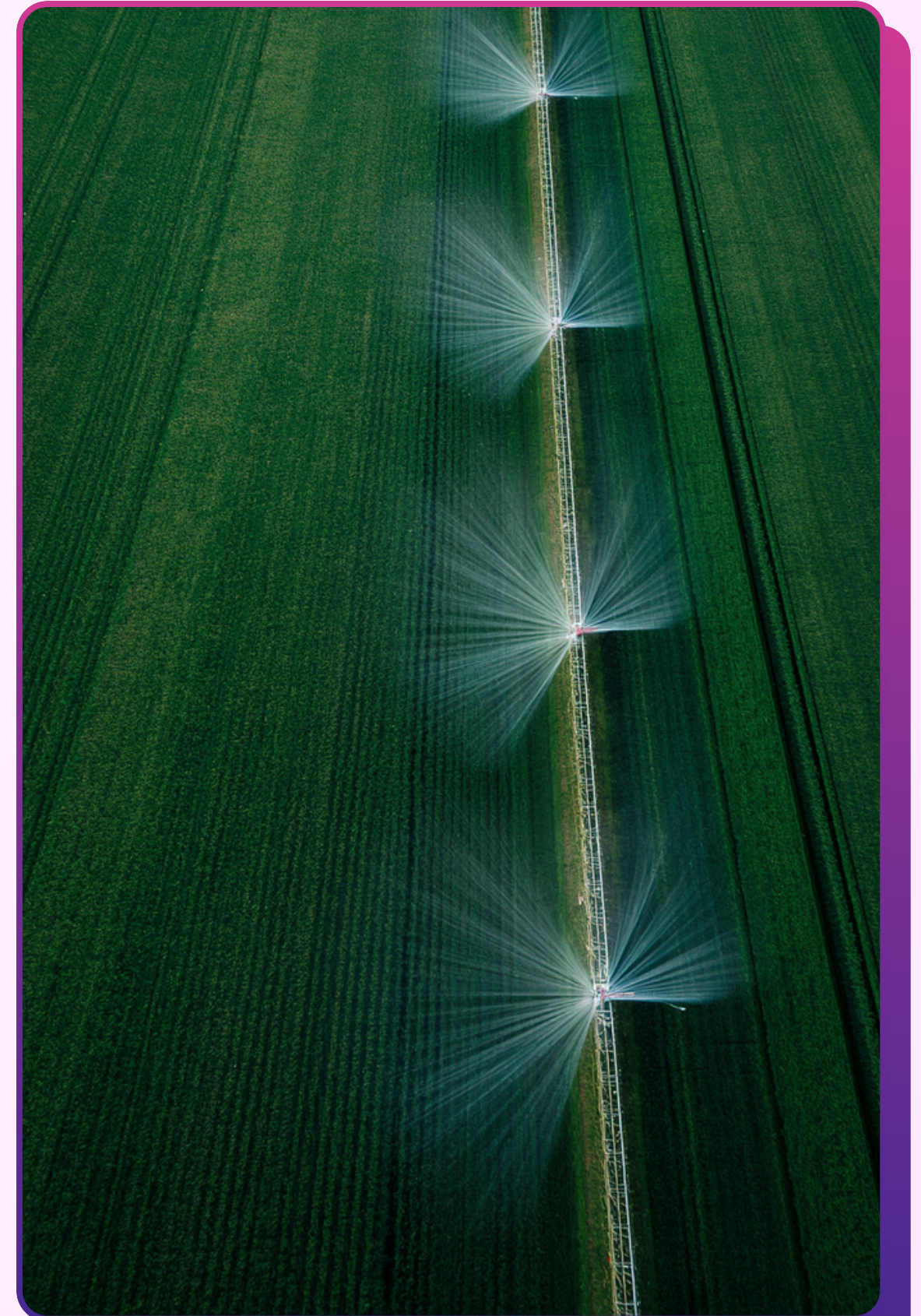
Traditional irrigation ignores real-time weather & soil data

Fixed irrigation schedules waste water and reduce crop yields. Without real-time soil moisture or weather data, farmers over-irrigate or under-irrigate.

HydroSense Kenya introduces a prediction step:

Weather Data → Soil Moisture → Prediction → Irrigation Decision

By modeling evapotranspiration and simulating soil dynamics, HydroSense enables data-driven, efficient irrigation management.



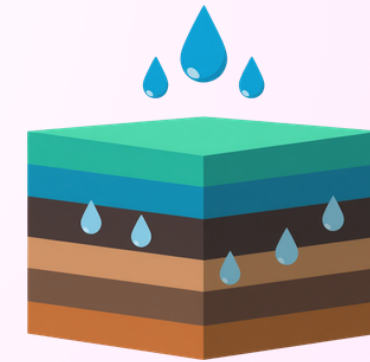
Project Objectives



Process Agricultural Datasets



Model Evapotranspiration



Simulate Soil Moisture

Project Objectives

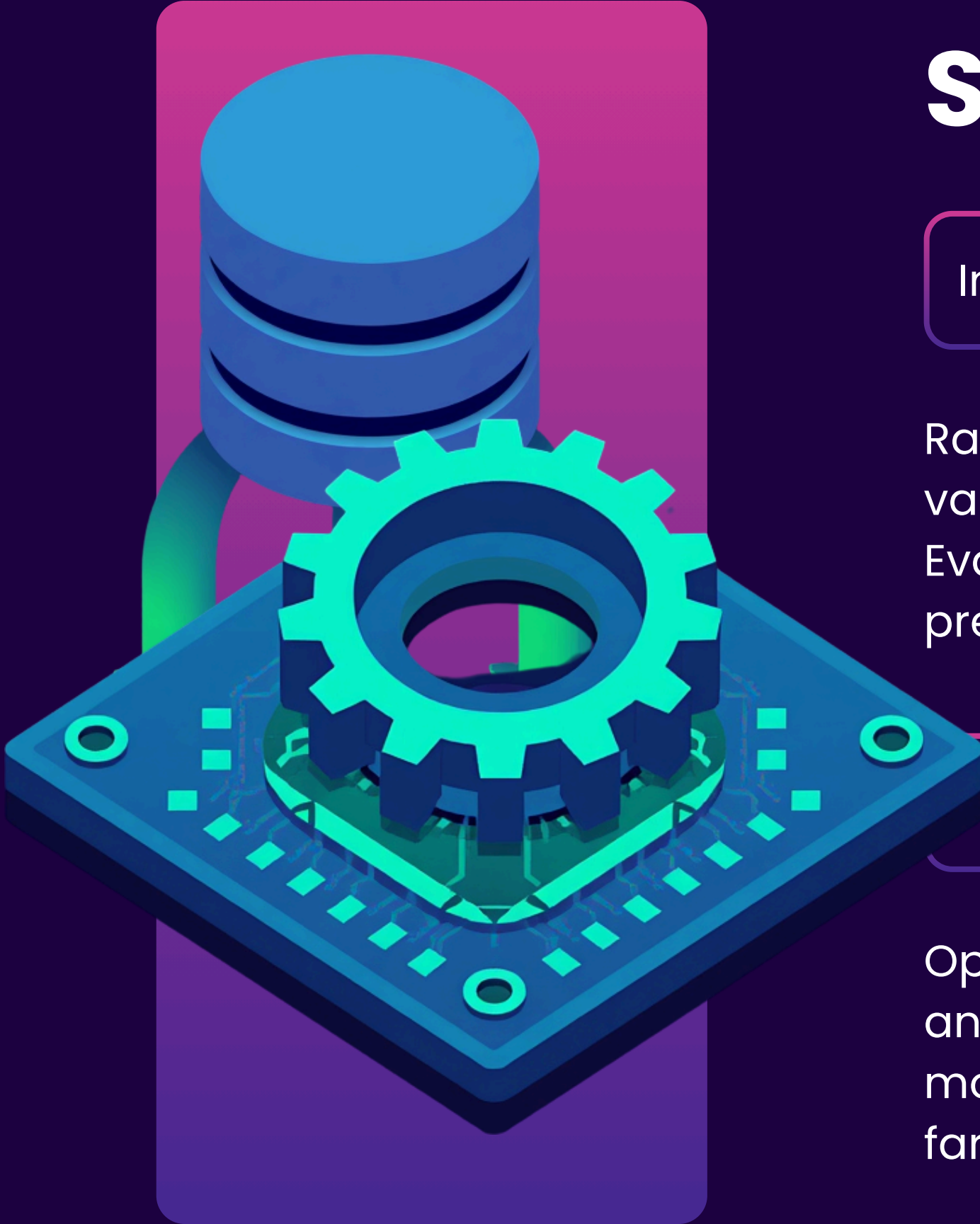
Predict Rainfall



Optimize Irrigation Schedules

Improve Water-Use Efficiency



An illustration on the left side of the slide featuring a blue database cylinder, a red gear, and a green circuit board with various components and solder points.

System Architecture

Inputs: Weather Data → Soil Data → Crop Data → Data Cleaning

Raw sensor and field data flows through automated cleaning and validation before entering the scientific computing engine. Evapotranspiration models, soil moisture simulations, and rainfall predictions are computed using Euler and RK4 numerical methods.

Pipeline: Scientific Computing → Numerical Simulation → Monte Carlo

Optimized outputs — irrigation schedules, water-use efficiency reports, and zone-level decisions — are delivered to the Decision Support module, enabling precise, data-driven irrigation management across all farm zones.

Datasets



Weather Dataset

Rainfall · Temperature · Humidity · Wind · Solar Radiation



Soil Dataset

Soil Moisture · Sensor Readings



Crop Dataset

Moisture Thresholds · Field Capacity · Drainage Coefficient

Water Balance Model

$$\Delta S = \text{Rainfall} + \text{Irrigation} - \text{ET} - \text{Drainage}$$

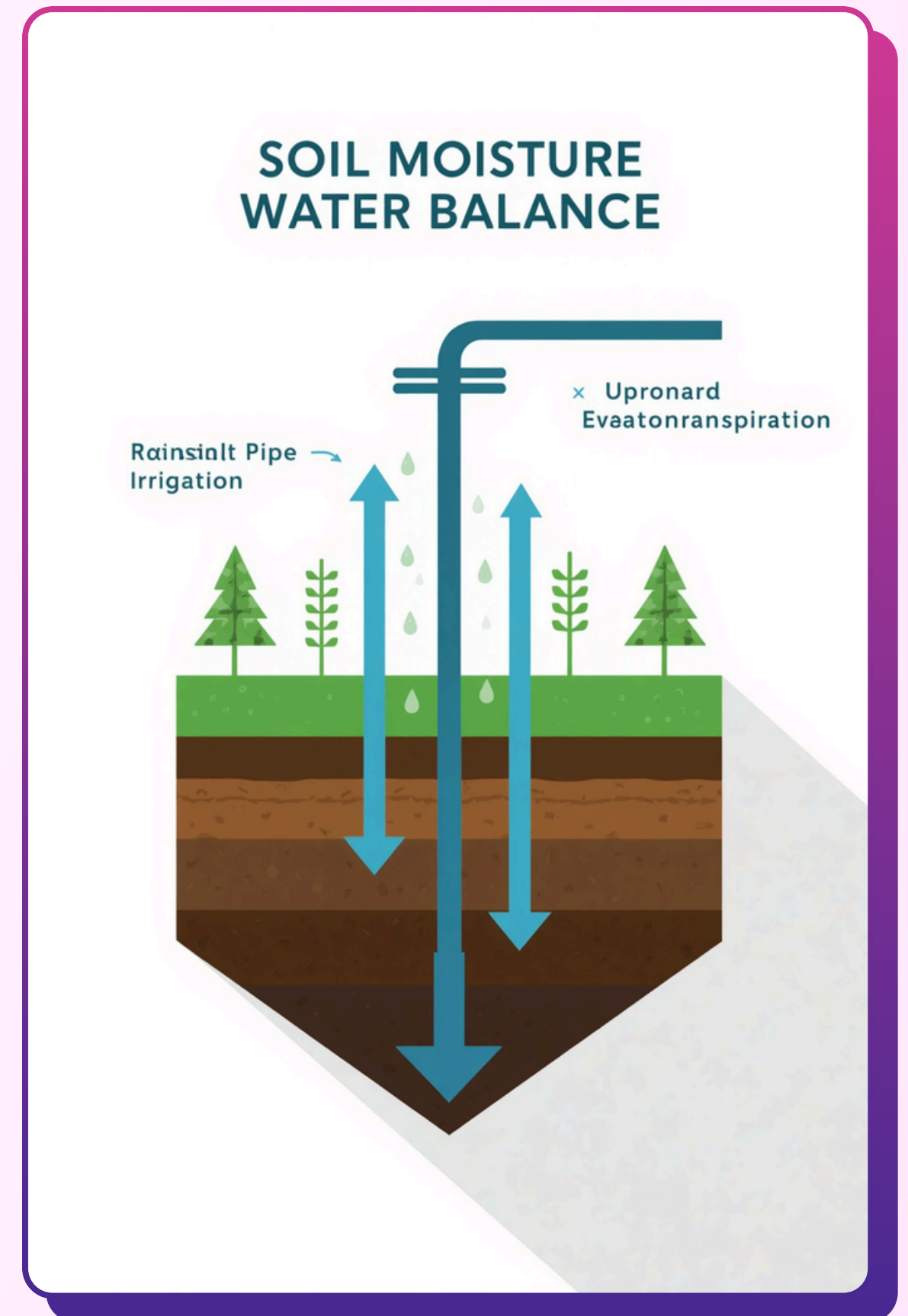
ΔS — Change in soil moisture storage

Rainfall — Natural precipitation input

Irrigation — Applied water from scheduling

ET — Evapotranspiration (crop water use)

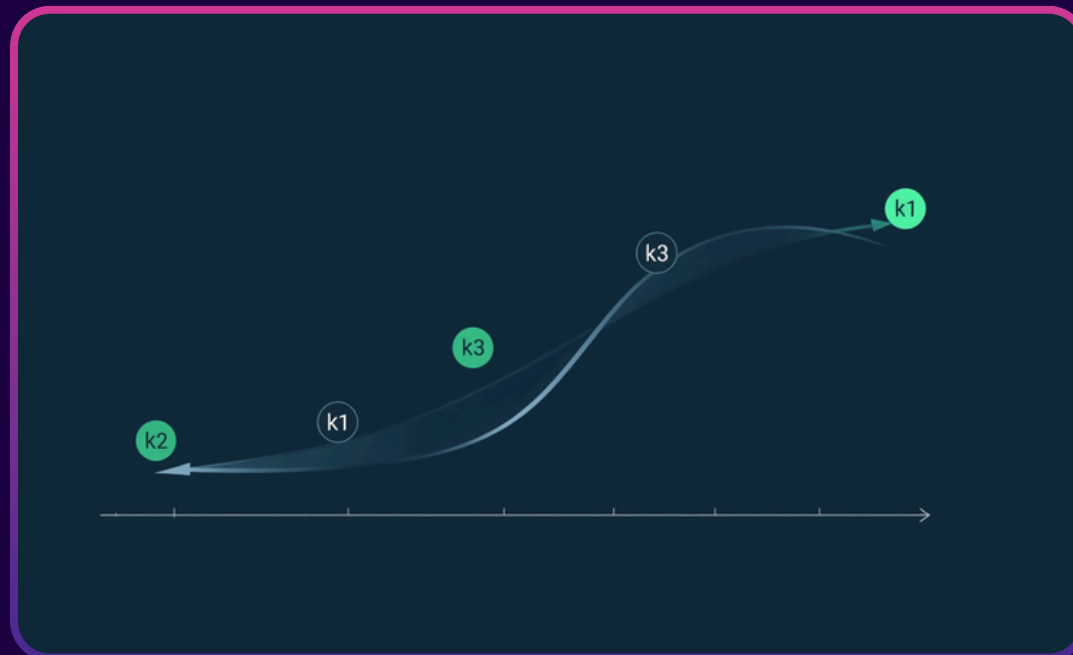
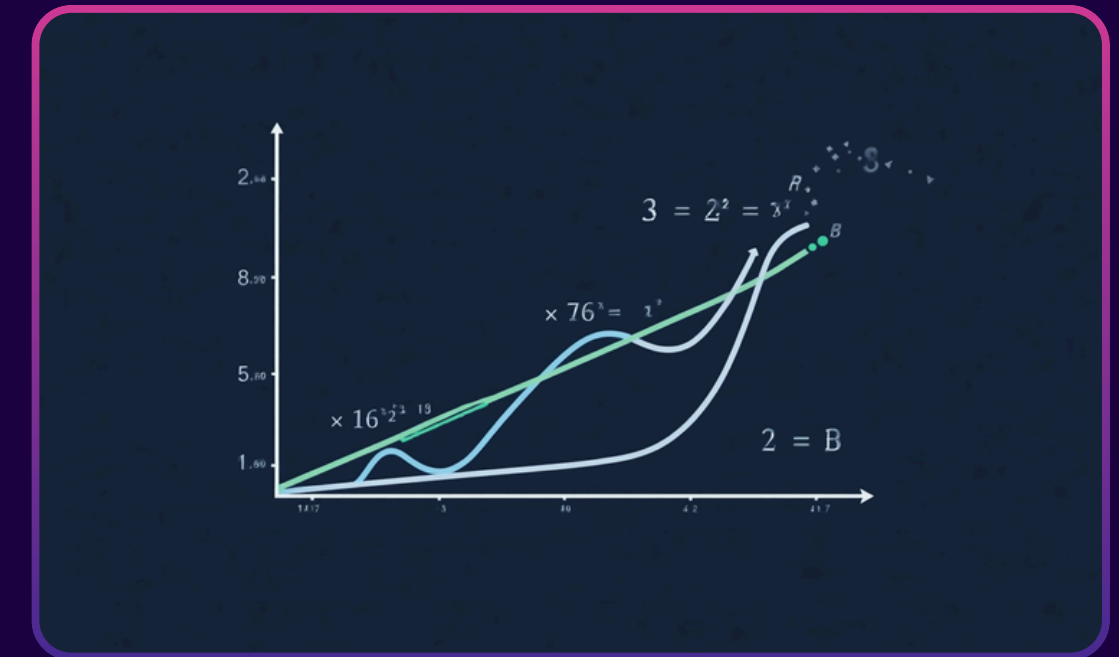
Drainage — Excess water leaving the root zone



Numerical Methods

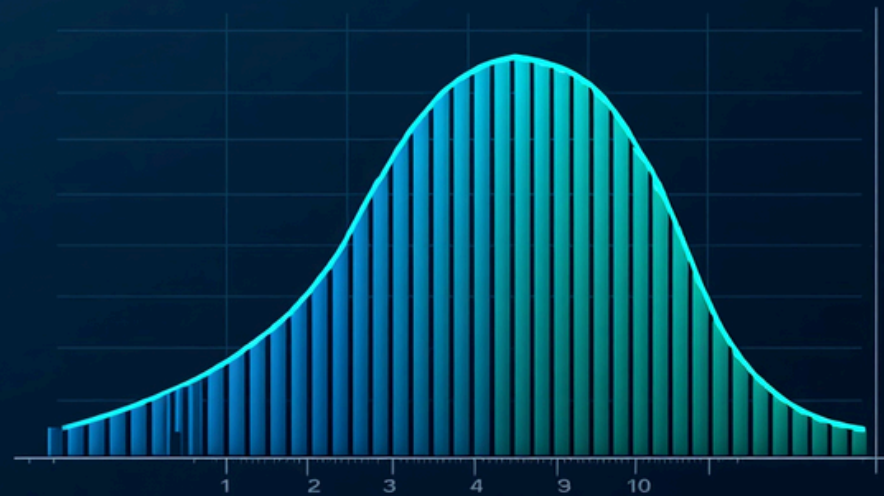
Euler Method — Simple & Fast

- ◆ Simple first-order approximation
- ◆ Fast computation per time step
- ◆ Lower numerical accuracy
- ◆ Suitable for rough estimates
- ◆ $\Delta S \approx S(t) + h \cdot f(t, S)$



Runge-Kutta (RK4) — Preferred Method

- ◆ Fourth-order accuracy — highly stable
- ◆ Uses four slope estimates per step
- ◆ Preferred for soil moisture simulation
- ◆ Handles non-linear dynamics well
- ◆ Daily simulation: $k1 \rightarrow k2 \rightarrow k3 \rightarrow k4 \rightarrow S(t+1)$



Monte Carlo Simulation

Historical Rainfall → Probability Distribution

Random Sampling → 1000 Simulated Scenarios

Risk-Aware Irrigation Decisions

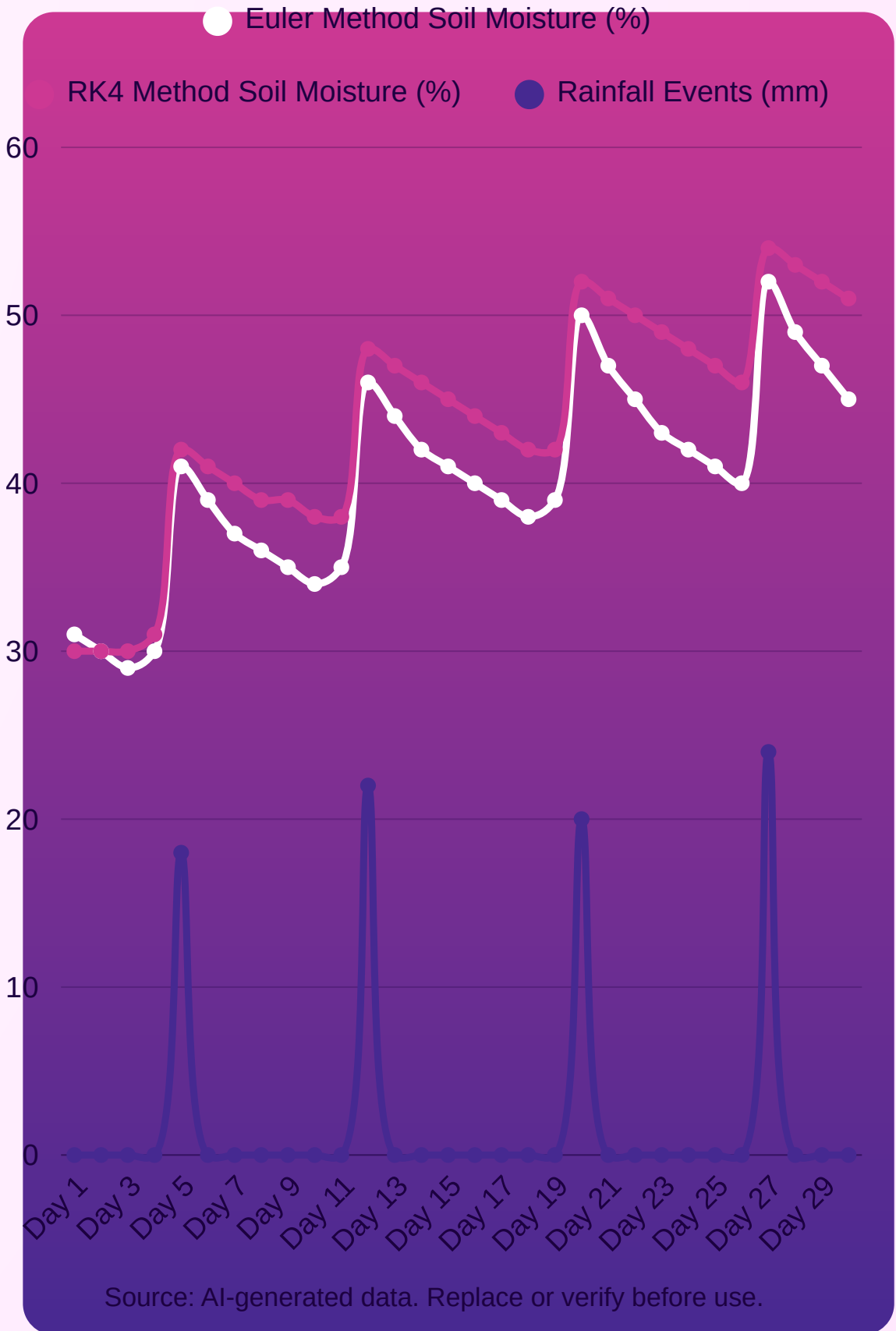
Simulation Results

Euler vs RK4 Soil Moisture Comparison

📈 Stable Prediction: RK4 maintains consistent soil moisture estimates with minimal drift over the 30-day simulation period.

💧 Rainfall Response: Both methods respond to rainfall events, but RK4 captures peak moisture levels more accurately.

📊 Moisture Trends: RK4 outperforms Euler in tracking gradual drainage and evapotranspiration cycles.





Intelligent Irrigation Scheduling

Weather → Predict Moisture → Compare with Threshold

Real-time weather data drives soil moisture predictions. If predicted moisture falls below the crop threshold, irrigation is triggered automatically — eliminating waste and ensuring optimal crop hydration at every growth stage.

Apply Irrigation Only When Necessary

Zone-level scheduling integrates RK4 simulation outputs with Monte Carlo rainfall forecasts to generate precise, risk-aware irrigation plans. Water is applied only when soil moisture deficits are confirmed, maximizing water-use efficiency across all farm zones.



Multi-Zone Optimization

Three farm zones feed into a central optimization engine to generate an intelligent Farm Irrigation Plan.

Zone A, Zone B, and Zone C data are processed through the optimization engine. Each zone receives a tailored irrigation schedule based on soil moisture, crop type, and water availability — minimizing waste and maximizing yield efficiency across the entire farm.



Testing & Validation

✓ Data preprocessing ✓ Water balance ✓ ET calculation

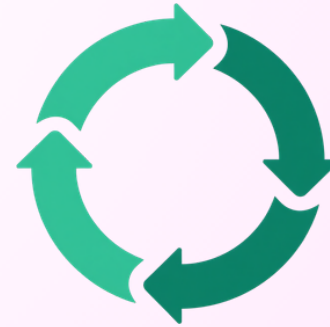
✓ Drainage model ✓ Euler simulation ✓ RK4 simulation

✓ Monte Carlo simulation ✓ Irrigation scheduling ✓ Multi-zone optimization

Project Achievements



Data Processing & Modeling



Simulation & Optimization

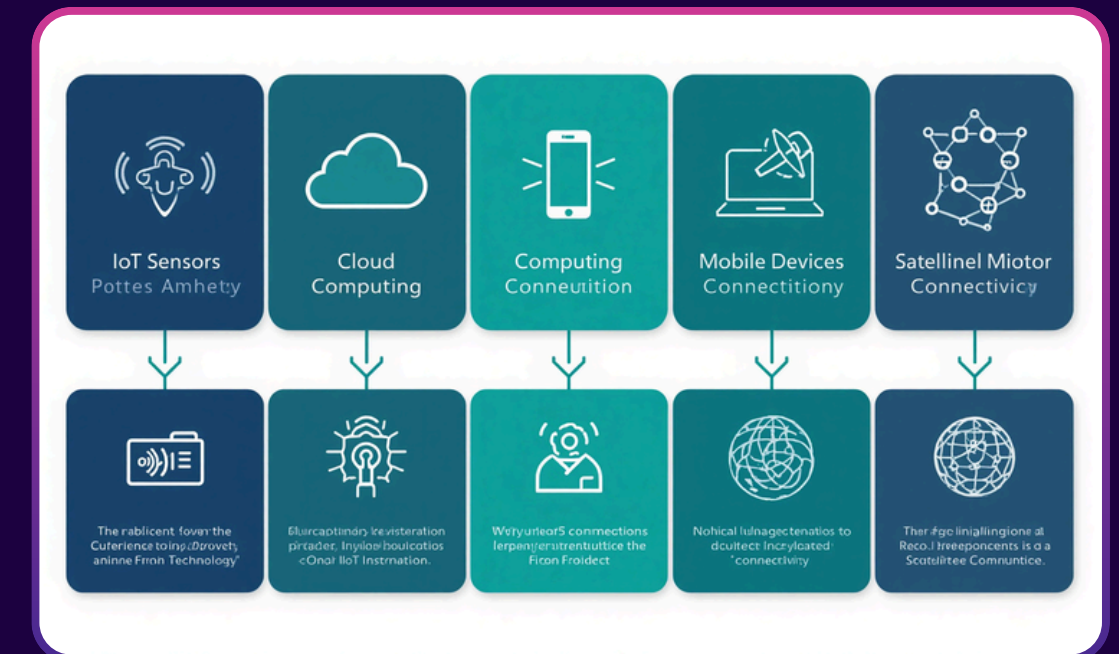


Decision Support System

Limitations & Future Work

Current Limitations

- Relies on historical data only
- No real-time IoT sensor input
- Limited to simulated scenarios
- Single-crop model assumption
- Manual data entry required



Future Improvements

- IoT Sensors — real-time soil data
- Machine Learning — adaptive models
- Mobile App — farmer interface
- Cloud Dashboard — remote monitoring
- Satellite Integration — field mapping
- Drone Monitoring — aerial analytics

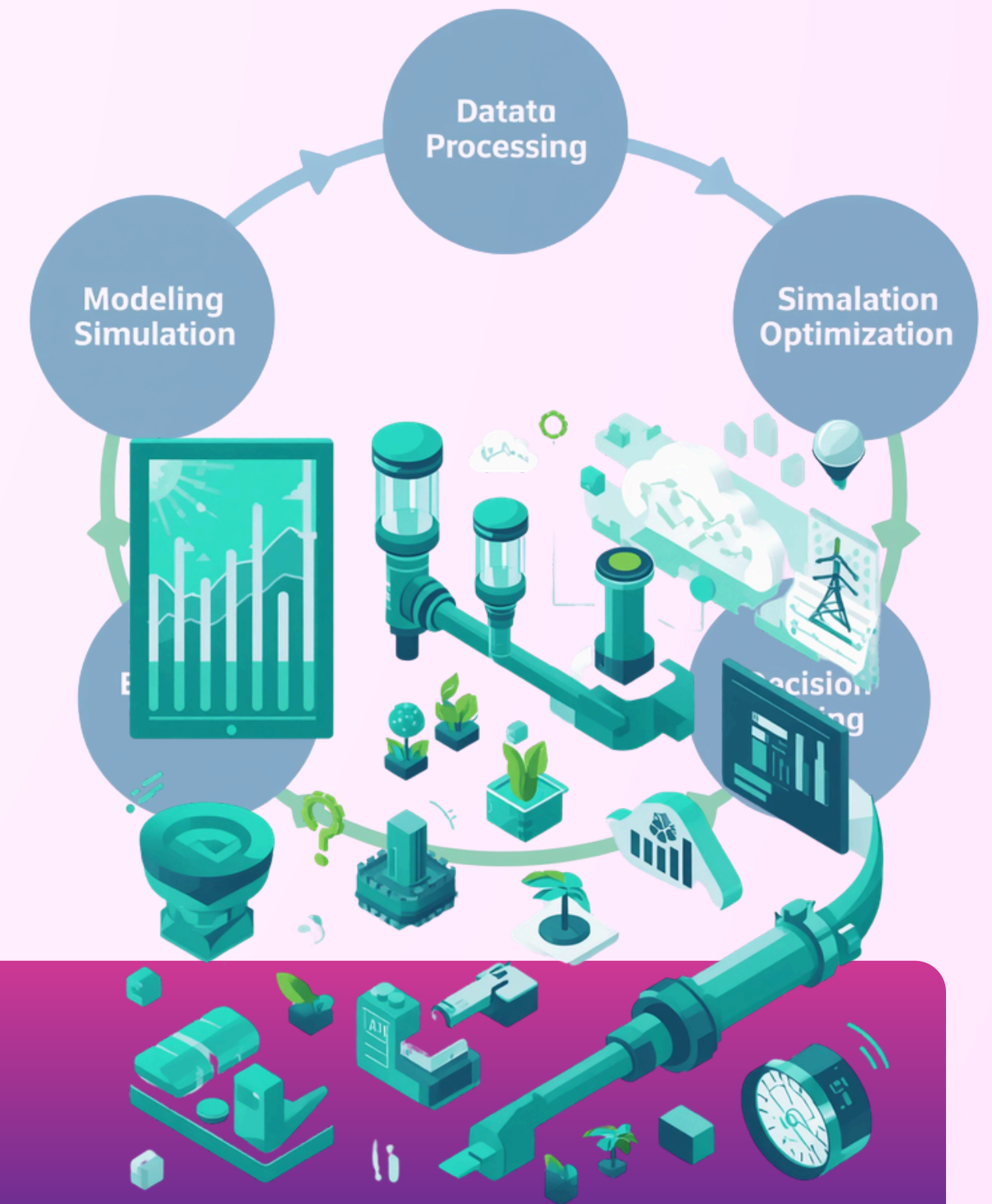


Conclusion

Scientific Computing transforms agricultural data into intelligent irrigation decisions.

Five Key Contributions:

- 1.Data Processing — Clean, validated agricultural datasets
- 2.Modeling — Water balance & evapotranspiration equations
- 3.Simulation — Euler, RK4 & Monte Carlo methods
- 4.Optimization — Multi-zone intelligent irrigation scheduling
- 5.Decision Making — Risk-aware, data-driven irrigation decisions





Questions?

**Scientific Computing for Intelligent
Irrigation Management**
Kenneth Mweu
SCT 211-0383/2024
Supervisor: Dr. Nderu

Thank You for Your Attention